

Vijaybhoomi School of Science & Technology

END-TERM EXAMINATIONS MAY 2026

Course Name: Machine Learning & Algorithms 2

Course Code: STAI205

Duration: 1.15 Minutes

Max. Marks: 40

Instructions to the students:

1. Attempt all questions.
2. Submit your results on the LMS according to the submission requirements, and appear for a viva-voce according to the schedule communicated to you separately.

Q. No.	Question	CO. No.	Marks
1	Your objective is to independently engineer, formulate, and solve three distinct machine learning problems using <i>Assessing Mathematics Learning in Higher Education</i> dataset sourced from the UCI Machine Learning Repository: https://shorturl.at/nWPjI preprocess the data and engineer appropriate features to formulate three separate analytical tasks: 1. Clustering: Define a methodology to identify underlying cohorts or patterns within the student data. 2. Classification: Define a predictive task with a discrete target variable. 3. Regression: Define a predictive task with a continuous target variable. For each of the three formulated problems: 1. Implement a baseline model to establish preliminary performance. 2. Implement an advanced Ensemble Method (e.g., AdaBoost, Gradient Boosting, XGBoost, or Stacking) to solve the task. 3. Execute and document a rigorous hyperparameter tuning process to manage the bias-variance tradeoff. You are required to submit the following: - A comprehensive iPython Notebook that reports the final performance metrics and mathematically justifies why the selected ensemble method outperformed or underperformed relative to your baseline model for that specific data distribution. - A text file containing all the prompts that you provided to create your output, if you used AI. This involves all prompts used in all AI tools like ChatGPT, Gemini, Claude or any other AI assistance agents for coding / research.	CO1, CO2, CO3, CO4	40

The rubrics used for evaluation during the viva-voce are as follows:

1. **Problem Formulation & Feature Engineering (20%)**
2. **Algorithmic Intuition (Baselines) (20%)**
3. **Ensemble Mechanics & Hyperparameter Defense (20%)**
4. **Evaluation & Mathematical Justification (20%)**
5. **Code Ownership & AI Transparency (20%)**

"What's true of counterfeiting money should also be true of counterfeiting humans. If governments took decisive action to protect trust in money, it makes sense to take equally decisive measures to protect trust in humans. Prior to the rise of AI, one human could pretend to be another, and society punished such frauds. But society didn't bother to outlaw the creation of counterfeit humans, since the technology to do so didn't exist. Now that AI can pass itself off as human, it threatens to destroy trust between humans and to unravel the fabric of society."

- **Yuval Noah Harari, Nexus: A Brief History of Information Networks from the Stone Age to AI**

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